Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

High-Performance Computing Lab for CSE **2026**

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Report for Project 1

Due date: 09 March 2026, 18:00 (Europe/Zurich)

HPC Lab for CSE 2025 — Submission Instructions

Following these instructions is mandatory: Submissions that do not comply **will not be considered**.

- **Submission Platform**: Assignments must be submitted via SAM-UP.
- **Required Files**: Your submission must include:
  - **Report**: A PDF summarizing your results and observations, based on this LaTeX template.
    - File name: `project_number_lastname_firstname.pdf`
    - You are allowed to discuss all questions with anyone you like; however: (i) your submission must list anyone you discussed problems with and (ii) you must write up your submission independently.
  - **Source Code Archive**: A **compressed tar archive**, named `project_number_lastname_firstname.tar.gz`, containing:
    - All source code files.
    - Build files and scripts (e.g., `Makefile`, batch job scripts, plot scripts, etc). If you modified any provided files, ensure they still work as expected.
    - The full LaTeX source of your report including all needed files to build the report (figures, listings, etc).
  - **Size Limit**: The total submission (PDF + archive) must not exceed **25 MB**.
  - **File Naming Rules**: Use only `ASCII` characters (avoid spaces, special symbols, and non-English letters).
- **Grading Process**: The TAs will evaluate your project based on:
  - Your report write-up.
  - Your code implementation.
  - Benchmarking your code's performance.

1. Euler warm-up [10 points]

Question 1: What is the module system, and how is it used?
The module system (Lmod, a Lua-based environment modules implementation) is used on HPC systems to dynamically manage software environments. It allows users to load, unload, and switch between different software versions without manually modifying system paths.
Loading all software globally would lead to conflicts between different versions of compilers and libraries. The module system avoids this problem by dynamically modifying environment variables.
In practice, before compiling or running programs on Euler, users load the required modules (e.g., compilers or libraries) inside their batch scripts using commands such as `module load`.

Question 2: What is Slurm, and what is its purpose?
Slurm is an open-source workload manager and job scheduler for Linux and Unix-like systems, widely used by supercomputers and clusters. It schedules and executes jobs on compute nodes, handling resource allocation, queuing, and execution control.
In particular (from the Slurm documentation), it provides three main functions:

- it allocates exclusive and/or non-exclusive access to resources (compute nodes, CPUs, memory, GPUs) to users for a specified duration of time so they can perform work;
- it provides a framework for starting, executing, and monitoring work (normally a parallel job) on the set of allocated nodes.
- it arbitrates contention for resources by managing a queue of pending jobs.

We now illustrate how the remaning questions were adressed.
The solution to question 3 and 4 can be found by first running the Makefile provided and then running the file `slurm_job_one.sh`. We already provide the ouputs of the batch script as requested. The output is reported also here for completeness:

```
Currently Loaded Modules:
1) stack/2025-06   2) gcc/12.2.0



Model name:                              AMD EPYC 9654 96-Core Processor
eu-g9-101-4
Hello World from eu-g9-101-4
```

As we can see we have specified to use a node with AMD EPYC 9654 96-Core Processor, and the job was executed on the node `eu-g9-101-4`.

The solution to question 5 can be found by running the file `slurm_job_two.sh`. The output of the batch script is reported here for completeness:

```
Currently Loaded Modules:
1) stack/2025-06   2) gcc/12.2.0



Model name:                              AMD EPYC 7763 64-Core Processor
eu-a2p-280
eu-a2p-285
Hello World from eu-a2p-285
Hello World from eu-a2p-280
```

2. Performance characteristics [50 points]

2.1. Peak performance [16 Points]

Proceeding as in the example given in the project description, the hardware specifications can be obtained from the Euler webpage.
For Euler VII (phase 1), the system consists of $n_{\text{nodes}} = 292$ nodes, each equipped with $n_{\text{sockets}} = 2$ sockets and $n_{\text{cores}} = 64$ cores per socket, operating at a frequency $f = 2.6\,\text{GHz}$.
For Euler VII (phase 2), the system consists of $n_{\text{nodes}} = 248$ nodes, each with $n_{\text{sockets}} = 2$ sockets and $n_{\text{cores}} = 64$ cores per socket, operating at a frequency $f = 2.45\,\text{GHz}$.

Using the uops.info database provided, we determine both the superscalability factor and the FMA factor for the corresponding AMD architectures.
For both Euler VII phases, which are based on AMD Zen2 and AMD Zen3 respectively, we obtain $n_{\text{super}} = 4$ and $n_{\text{FMA}} = 2$.

The FMA factor is determined by observing that all instructions marked as **FMA** in the table have a corresponding number of micro-operations equal to one. This indicates that the fused multiply-add is executed as a single micro-operation by dedicated hardware. Since an FMA instruction performs the operation $a \cdot b + c$, it consists of one floating-point multiplication and one floating-point addition. Therefore, each FMA instruction corresponds to two floating-point operations, and thus $n_{\text{FMA}} = 2$.

The superscalability is defined as the number of floating-point operations that the FPU of a core can execute in parallel within a single clock cycle. To determine this value, we analyze the instruction `VFMADD132PD (YMM, YMM, YMM)` reported in the uops.info table. The table specifies a throughput of 0.50 cycles per instruction, one micro-operation, and execution on ports FP0/1. A throughput of 0.50 cycles per instruction implies that

$$\frac{1}{0.50} = 2$$

FMA instructions can be issued per cycle. The port information FP0/1 confirms the presence of two independent floating-point execution pipelines capable of executing FMA instructions in parallel.

Since two FMA instructions can be executed per cycle and each FMA instruction corresponds to two floating-point operations, the total number of floating-point operations per cycle is

$$2 \times 2 = 4.$$

Hence, for both AMD Zen 2 and AMD Zen 3, the superscalability is

4 floating-point operations per cycle per core.

To determine n_{SIMD}, we inspect the floating-point instructions reported in the uops.info table, where instructions such as `VFMADD132PD (YMM, YMM, YMM)` appear. The presence of YMM operands indicates the use of 256-bit AVX registers, since YMM registers are defined as 256-bit wide in the AVX/AVX2 instruction set. As both AMD Zen 2 and Zen 3 execute these YMM-based instructions, they necessarily support AVX2 with 256-bit SIMD registers. A double-precision floating-point number occupies 64 bits; therefore, the number of doubles that can be processed concurrently by one SIMD instruction is

$$n_{\text{SIMD}} = \frac{256}{64} = 4.$$

Hence, each SIMD instruction can operate on four double-precision values in parallel.

Parameter	**Euler VII (Phase 1)**	**Euler VII (Phase 2)**
n_{nodes}	292	248
n_{sockets}	2	2
n_{cores} (per socket)	64	64
Clock frequency f	2.6 GHz	2.45 GHz
n_{FMA}	2	2
n_{super} (FP ops/cycle/core)	4	4
n_{SIMD} (doubles per SIMD instr.)	4	4

Table 1: Summary of hardware parameters and architectural factors used for peak performance estimation.

We can now compute all the peak performance parameters for both Euler VII phases :

$$P_{\text{core},1} = n_{\text{super}} \times n_{\text{FMA}} \times n_{\text{SIMD}} \times f = 4 \times 2 \times 4 \times 2.6\,\text{GHz} = 83.2\,\text{GFLOPS/core}.$$
$$P_{\text{CPU},1} = n_{\text{cores}} \times P_{\text{core},1} = 64 \times 83.2\,\text{GFLOPS/core} = 5.3248\,\text{TFLOPS/CPU}.$$
$$P_{\text{node},1} = n_{\text{sockets}} \times P_{\text{CPU},1} = 2 \times 5.3248\,\text{TFLOPS/node} = 10.6496\,\text{TFLOPS/node}.$$
$$P_{\text{cluster},1} = n_{\text{nodes}} \times P_{\text{node},1} = 292 \times 10.6496\,\text{TFLOPS/node} = 3110.08\,\text{TFLOPS}.$$

$$P_{\text{core},2} = n_{\text{super}} \times n_{\text{FMA}} \times n_{\text{SIMD}} \times f = 4 \times 2 \times 4 \times 2.45\,\text{GHz} = 78.4\,\text{GFLOPS/core}.$$
$$P_{\text{CPU},2} = n_{\text{cores}} \times P_{\text{core},2} = 64 \times 78.4\,\text{GFLOPS/core} = 5.017\,\text{TFLOPS/CPU}.$$
$$P_{\text{node},2} = n_{\text{sockets}} \times P_{\text{CPU},2} = 2 \times 5.017\,\text{TFLOPS/node} = 10.034\,\text{TFLOPS/node}.$$
$$P_{\text{cluster},2} = n_{\text{nodes}} \times P_{\text{node},2} = 248 \times 10.034\,\text{TFLOPS/node} = 2488.432\,\text{TFLOPS}.$$

2.2. Memory Hierarchies [8 Points]

2.2.1. Cache and main memory size

We report in table the results. We do not report here the ouputs. One can find the in the given files of the repo

Main memory (total)	249 GB
L3 cache (total)	512 MB (32 × 16 MB)
L3 cache (per socket)	256 MB (16 × 16 MB)
L2 cache (per core)	512 KB
L1 data cache (per core)	32 KB
L1 instruction cache (per core)	32 KB

Table 2: Memory hierarchy of an Euler compute node with AMD EPYC 7H12 (2 sockets, 64 cores per socket). Each core has private L1 and L2 caches, while L3 cache is organized in 32 blocks of 16 MB shared among groups of 4 cores.

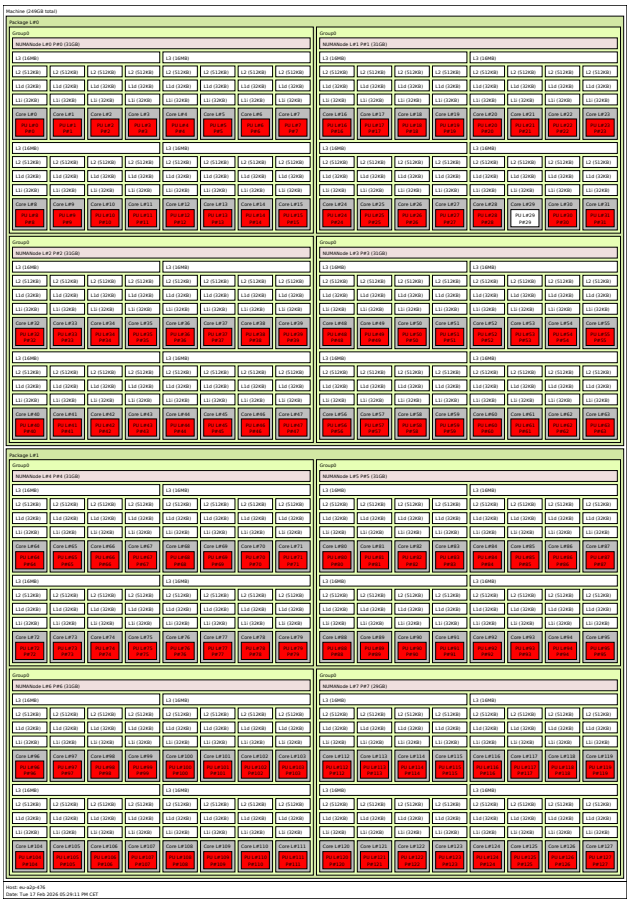


Figure 1: Topology of Euler VII (phase 1) compute node with AMD EPYC 7H12 (2 sockets, 64 cores per socket).

Main memory (total)	249 GB
L3 cache (per socket)	256 MB (8 × 32 MB)
L3 cache (total)	512 MB (16 × 32 MB)
L2 cache (per core)	512 KB
L1 data cache (per core)	32 KB
L1 instruction cache (per core)	32 KB

Table 3: Memory hierarchy of an Euler compute node with AMD EPYC 7763 (2 sockets, 64 cores per socket). Each core has private L1 and L2 caches, while the L3 cache is organized in 16 blocks of 32 MB shared among groups of cores.

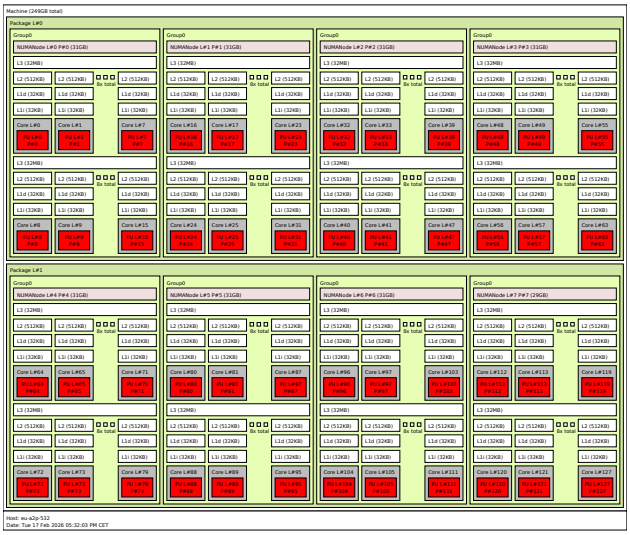


Figure 2: Topology of an Euler compute node with AMD EPYC 7763 (2 sockets, 64 cores per socket).

2.3. Bandwidth: STREAM benchmark [10 Points]

2.4. Performance model: A simple roofline model [16 Points]

3. Auto-vectorization [10 points]

Question 1: what is data alignment and is it important for data structures to be aligned?

Question 2: what are some obstacles that can prevent automatic vectorization by the compiler?

Question 3: is there a way to help the compiler to vectorize and how?

Question 4: which loop optimizations are performed by the compiler to vectorize and pipeline loops?

Question 5: what can be done if automatic vectorization by the compiler fails or is suboptimal?

4. Matrix multiplication optimization [30 points]